Ad Performance Intelligence Engine

ML + RAG system that detects anomalies, forecasts ROAS, and explains WHY your campaign broke

Viktoria Repich

Growth Leader Who Codes | 13+ years healthtech, fintech, D2C

The Problem

$2K+/month

SMBs pay for dashboards that show numbers they already know

24.8% data loss

GA4 misses a quarter of touchpoints due to cookie rejection

Last-click lies

Google Brand gets 20.7% credit. Shapley says it deserves 15.2%

No WHY

Looker tells you ROAS dropped. But never explains the cause

The Solution

7 ML Models

Auto-selects best model.
Anomaly F1=0.91, ROAS R^2=0.90

SHAP Explainability

Not just 'anomaly' — but 'CTR dropped 35%, CPC spiked 50%'

RAG + LLM

Ask in plain English.
Answers from YOUR data, not hallucinations

4 Platform Connectors

Google Ads, Meta, TikTok, GA4.
Unified format, one pipeline

5 Attribution Models

Shapley-based. Shows which channel actually caused the sale

143 Tests + Docker

Production-ready.
Not a Jupyter notebook

ML Model Results

Anomaly Detection

Model	F1
Random Forest 🏆	**0.91**
XGBoost	0.91
Logistic Regression	0.81
Isolation Forest	0.47

ROAS Forecasting

Model	MAE	R^2
LightGBM 🏆	**0.18**	**0.90**
XGBoost	0.21	0.84
Linear Regression	0.25	0.83

0.91
Best F1 Score

0.90
Best R^2

143
Pytest Tests

Attribution: Last-Click Lies

Channel	Last-Click	Shapley	Verdict
TikTok Paid	10.9%	14.5%	UNDERVALUED
Meta Facebook	11.0%	14.3%	UNDERVALUED
Google Brand	20.7%	15.2%	OVERVALUED
Direct	12.1%	11.2%	Fair

Cookieless Tracking Recovery

24.8%
GA4 data loss

22.3%
Recovered by enhanced tracking

97.5%
Total visibility achieved

Tech Stack

Layer	Tools
Data & ETL	pandas, numpy, 25 engineered features
ML Models	scikit-learn, XGBoost, LightGBM, SHAP
RAG	ChromaDB, sentence-transformers, Llama 3.1
Attribution	Custom Shapley, cookieless tracking sim
API	FastAPI (8 endpoints), uvicorn
Dashboard	Streamlit (6 pages), Plotly
Integration	MCP server (6 tools for Claude Desktop)
Connectors	Google Ads, Meta Ads, TikTok Ads, GA4
DevOps	Docker, MLflow, pytest (143 tests)

Roadmap

COMPLETED

- ✅ ETL pipeline + 25 features
- ✅ 7 ML models + SHAP explainability
- ✅ RAG (ChromaDB + Llama 3.1)
- ✅ FastAPI + Streamlit + MCP server
- ✅ 4 platform connectors
- ✅ Multi-touch attribution (5 models)
- ✅ MLflow + Docker

NEXT

- ☐ Real API integrations (live ad accounts)
- ☐ PostHog for production attribution
- ☐ Self-serve product for SMBs

Let's Talk

Looking for senior marketers, growth leads, and business owners for coffee chats and user interviews

GitHub

github.com/vikarepich

LinkedIn

linkedin.com/in/repich-viktoriia/

No pitch. Just your pain points, your workflow, your 'I wish this tool existed' moment.